

Performance evaluation of Component Based Software¹

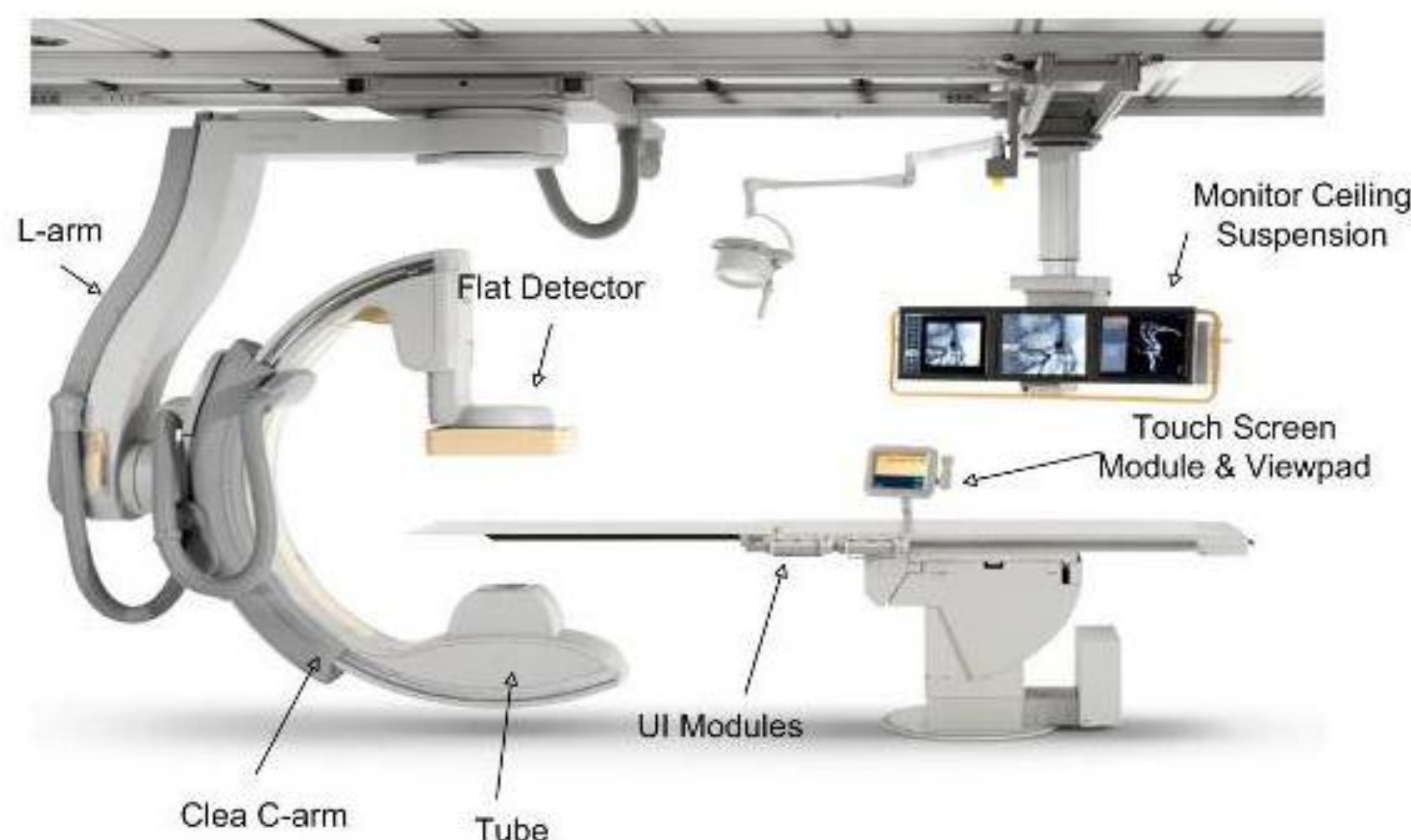
By Freek van den Berg, PhD candidate Performance Evaluation of Component-Based Software

Based on a Interventional X-ray machine by Philips Healthcare, performance evaluation on component based software is performed. The results obtained are used to create new methodologies for performance analysis, which, in turn, are tested in the Philips environment.

Many companies developing high-tech products would like to have a higher rate of innovation. Basically because they need to market products quickly and cost effectively, without compromising the high standards of safety, performance and reliability.

The Allegio project works closely with the Philips Healthcare iXR Group to develop and evaluate architecting methodologies for effective, high-tech systems, including the interventional X-ray machines

The interventional X-ray (iXR) machines, named Allura, can be used for performing minimal-invasive (cardiac, vascular and neuro) surgeries and consist of several moving parts.



Conducting performance evaluation at an early stage, can prevent making costly mistakes in the development, support development decisions and lead to more confidence in the quality of the product.

The research project

The cooperation with Philips will last four years and provides a couple of case studies. The cases involve the Allura with access to its software. Performance evaluation on this software will be the area of concern.

The first year of my PhD project will cover two software components of the Allura, namely collision prevention and distance computations. Key points are worst case situations in order to ensure safety, performance and reliability at all times.

In the remaining three years more components of the Allura and their interactions will be researched. This results into a generic toolbox for performance evaluation on software that matures over time.

The Allura case study

The Allura contains fast-moving, big parts. High quality software is needed to exclude the probability of a doctor, patient or nurse getting injured. This is done using collision prevention.

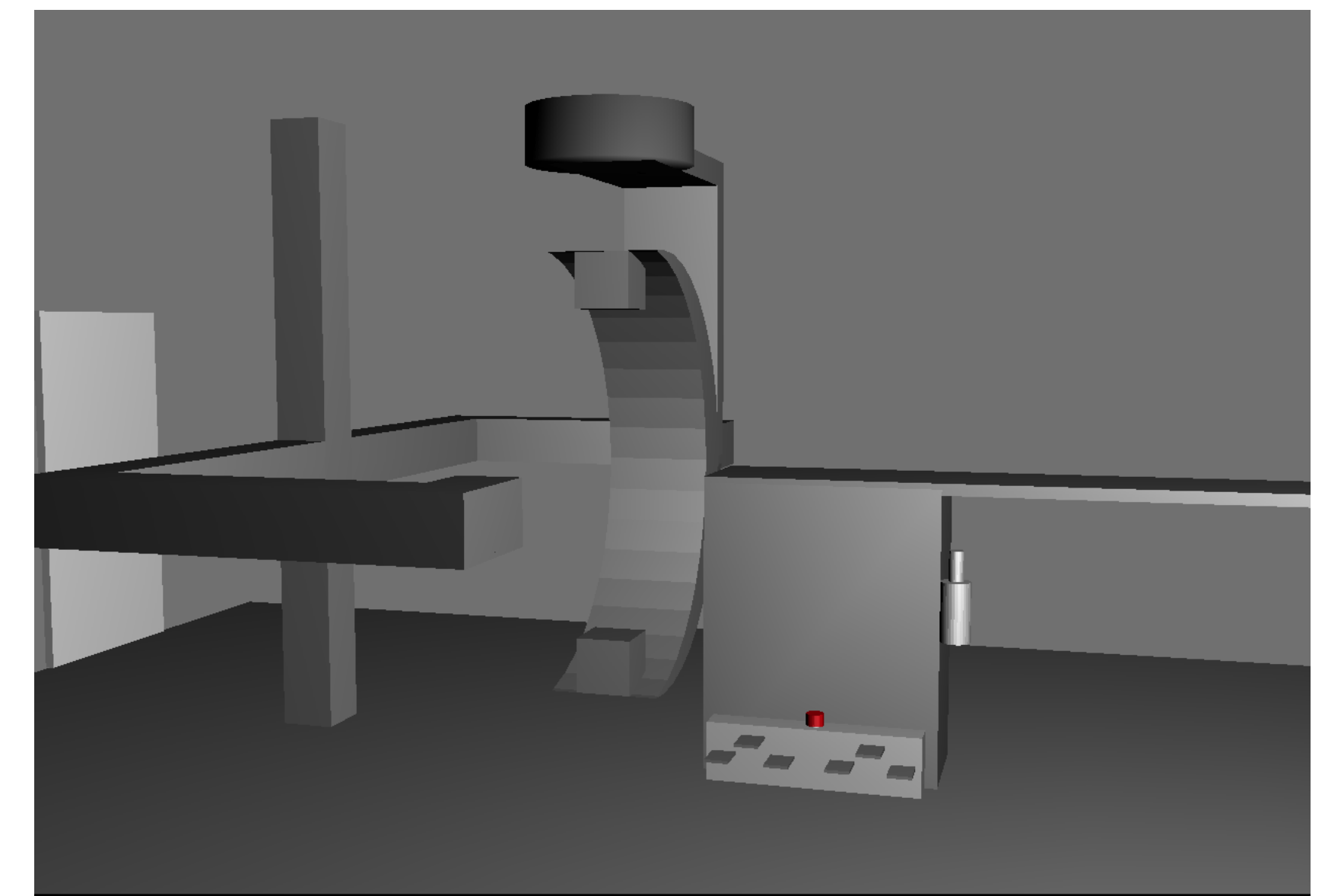
Collision prevention can be done using three layered levels:

- Geometric positioning
- Collision prevention
- Distance computations

Geometric positioning

The first layer, named geometric positioning (GP), contains a geometric model of the system. The model is updated using sensor and motor information.

Human operated UI controls (combined with collision prevention) dictate what the system will actuate, in a safe way.



Collision prevention

At the second layer, the Collision prevention on, the GP model is analyzed for collision danger.

One cardinal aspect of determining collisions is computing the distance between the different objects of the model on a frequent base. The distance computation is delegated to a function at the 3rd layer.

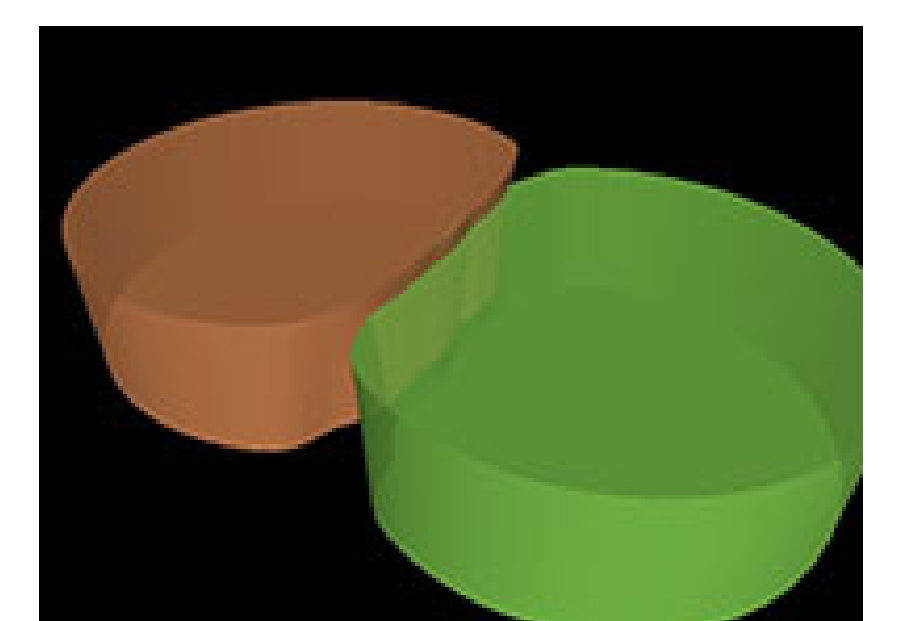
Using a high number of computed distances gathered over time, collision prevention can among others detect objects heading toward each other and use look-ahead to determine breaking distances.

Distance computations

At the third layer distance computations take place. A computation takes two objects from the model, computes and returns the distance between them.

To determine the safety, performance and reliability of the system, the frequency of and the speed at which the computations are dispatched matter much.

The frequency is defined as the number of computations per time interval. It is very dependent on the amount of stress at the collision prevention level.



The speed states how long a computation takes. It depends on factors as object complexities, software and hardware platform, and the mandated accuracy

Research goals

- Developing performance evaluation methods aiding the creation of new software components
- Developing performance evaluation methods for testing existing software components

Freek van den Berg (freek@vdberg.us)